THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS FULL SYLLABUS PAPER SET 01

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Which of the following is a perfect cube? 1

(a) 130 **(b)** 150 **(c)** 200 **(d)** 125

2. $\sqrt[3]{1728}$ is: 1

(a) 11 **(b)** 13 **(c)** 12 **(d)** 14

3. $3^{10} \div 3^{6}$ equals: 1

(a) 3^{4} **(b)** 3^{60} **(c)** 3^{1} **(d)** 3^{16}

4. The decimal number 13 written in base 2 is: 1

(a) 1110 **(b)** 1111 **(c)** 1100 **(d)** 1101

5. Three angles of a quadrilateral are 70°, 120° and 75°. The fourth angle is: 1

(a) 95° **(b)** 115° **(c)** 85° **(d)** 105°

6. Opposite sides of a parallelogram are: 1

(a) perpendicular **(b)** only parallel

(c) parallel and equal **(d)** only equal

7. Which of these is divisible by 5? 1

(a) 3472 **(b)** 3471 **(c)** 3475 **(d)** 3476

8. The value of $25 \times 4 \times 7$ using the best grouping is: 1

(a) 1400 **(b)** 70 **(c)** 700 **(d)** 350

9. 4.2 litres in mL is: 1

(a) 42000 mL **(b)** 42 mL **(c)** 4200 mL **(d)** 420 mL

10. The ratio of 250 g to 2 kg is: 1

(a) 1 : 8 **(b)** 5 : 4 **(c)** 1 : 80 **(d)** 1 : 4

11. The simple interest on ₹ 15000 at 6% per annum for 4 year(s) is: 1

(a) ₹ 3600 **(b)** ₹ 18600 **(c)** ₹ 1800 **(d)** ₹ 7200

12. 35% of 600 is: 1

(a) 220 **(b)** 420 **(c)** 200 **(d)** 210

13. If $a^2 + b^2 < c^2$ in a triangle, the triangle is: 1

(a) equilateral **(b)** acute-angled **(c)** obtuse-angled **(d)** right-angled

14. The distance between points (0, 0) and (5, 12) is: 1

(a) 17 **(b)** 12 **(c)** 7 **(d)** 13

15. If A : B = 2 : 3 and B : C = 3 : 5, then A : B : C is: 1

(a) 6 : 9 : 15 **(b)** 2 : 5 : 3 **(c)** 3 : 2 : 5 **(d)** 2 : 3 : 5

16. The number of squares in the net of a cube is: 1

(a) 8 **(b)** 12 **(c)** 6 **(d)** 4

17. The mean of the data 100, 120, 140, 160, 180 is: 1

(a) 140 **(b)** 138 **(c)** 141 **(d)** 142

18. The mean of n, $n+2$ and $n+4$ is: 1

(a) $n+2$ **(b)** $3n+6$ **(c)** $n+4$ **(d)** $n+6$

19. A rectangle has area 96 cm² and length 12 cm. Its breadth is: 1

(a) 8 cm **(b)** 7 cm **(c)** 84 cm **(d)** 10 cm

20. The unit of area is: 1

(a) units **(b)** square units **(c)** cubic units **(d)** none

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Which is greater, 2^{10} or 10^2? Why? 2

22. What is a tally mark system? 2

23. Find the remainder when 245 is divided by 9 using digit sum. 2

24. State two ways to check that two ratios are equal. 2

25. Find the area of a right triangle with hypotenuse 17 cm and one leg 8 cm. 2

26. Find the central angle of a sector representing 35% of the data. 2

27. How does adding 10 to every observation change the median? 2

28. What is the area of a triangle with vertices forming a right angle with legs 9 and 40? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Is 1372 a perfect cube? If not, by which smallest number should it be multiplied to make it one? 3

30. A square has diagonal 14 cm. Find its area using $\frac{1}{2}d^2$. 3

31. Expand and simplify $(x+2)(x+9)$. Verify for $x=1$. 3

32. At what total amount will ₹ 7500 grow in 3 years at 6% simple interest? 3

33. A cube has edge 6 cm. Find its surface area and the number of such cubes needed to make a cuboid of dimensions $12 \times 6 \times 6$. 3

34. Show that the difference of a two-digit number and its reverse is a multiple of 9. 3

35. Using $\left(n, \frac{n^2-1}{2}, \frac{n^2+1}{2}\right)$ find the triplet for $n=9$ and verify. 3

OR

Find the length of the diagonal of a rectangle whose perimeter is 34 cm and one side is 8 cm. 3

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36. A triangular field has base 18 m and height 12 m. Find its area and the cost of levelling at ₹ 12 per m². 3

OR

A path 2 m wide runs around a 20 m × 16 m rectangular garden outside it. Find the area of the path. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) Explain how to compare two numbers written in scientific notation.
(ii) Compare 4.5×10^7 and 9.1×10^6. 5

OR

(i) Write $\left(\frac{1}{2}\right)^{-5}$ without a negative exponent.
(ii) Simplify $\frac{(-2)^3 \times (-2)^{-5}}{(-2)^{-1}}$. 5

38. (i) How is GST calculated?
(ii) A restaurant bill is ₹ 1800 with 5% GST; find the total. 5

OR

Case study: A shop reduces the price of a school bag from ₹ 900 to ₹ 765. 5
(i) Find the discount.
(ii) Discount percent.
(iii) Find the price after a further 10% off.

39. **Case study:** A cricketer's runs in 8 matches: 12, 45, 0, 38, 62, 18, 45, 90. 5
(i) Find mean and median.
(ii) Find range.
(iii) Comment.

OR

(i) Describe how a line graph is read. 5
(ii) Name two real situations best shown by line graphs.

40. **Case study:** A kite-shaped garden has diagonals 40 m and 30 m perpendicular to each other. 5
(i) Find its area.
(ii) Compare with a rhombus of the same diagonals.
(iii) Cost of grass at ₹ 8 per m².

OR

(i) Rectangles of equal perimeter 40 cm can have different areas. Give four examples. 5
(ii) Which is largest?

END OF QUESTION PAPER Total: 80 Marks

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